Lakshmi Multispecialty Hospital Appointment System

Software tools used & step-by-step build guide

Virudhunagar, Tamil Nadu · Version of 16 October 2026 · Code: github.com/mnivi2021-commits/LakshmimulltiyDoctrappot2026

1. What the system does

Patients can book an appointment with Dr. Vijul (Orthopaedic), Dr. Vishwa (Diabetologist, Head of Hospital) or Dr. Nivi (Heart Specialist) in four ways:

- **AI phone call** – an AI receptionist (OmniDimension) answers in Tamil, English and Hindi, checks real free slots and books.
- **Website chat box** – a guided chat (name → mobile → doctor → date → morning/evening → time slot → confirm) that gives a booking reference number.
- **Book Appointment page / form** on the full website, served from the hospital PC.
- **Mobile app** (Expo Go) with AI CALL, CHAT BOX and BOOK APPOINTMENT buttons.

Staff see everything in the **admin dashboard** (PC) and in the Google Sheet **"Lakshmi Hospital – Calls & Requests"**, which keeps working even when the PC is switched off.

2. Software tools used

Tool	What it is used for	Cost
Building & editing		
Visual Studio Code	Code editor where all files were written and opened.	Free
Claude Code (AI assistant)	Wrote, tested and deployed the code from VS Code, following your instructions.	Subscription
Git + GitHub	Saves every version of the code online (the repository above). `git push` uploads changes.	Free
PowerShell / Git Bash	Windows command line used to run, test and deploy.	Free
Backend (runs on the hospital PC)		
Node.js 24 + npm	Runs the hospital server (`server.js`) and installs libraries.	Free
Express	Web server: website pages, booking API, voice-agent API, admin dashboard.	Free
XAMPP (MySQL / MariaDB)	Main database: doctors, schedules, appointments (LMH-0001 numbers), call logs. Port 3306.	Free
mysql2 · node:sqlite	Database drivers. SQLite is used only for the automatic tests.	Free

Tool	What it is used for	Cost
Node test runner	106 automatic tests (booking, voice, chat, Sheets, WhatsApp bot, pages). Run with `npm test`.	Free
cloudflared (Cloudflare Tunnel)	Gives the PC a temporary public https address so the AI phone agent can reach it.	Free
Always-on cloud parts (work when the PC is off)		
Google Sheets + Apps Script	Sheet "Lakshmi Hospital – Calls & Requests". `Code.gs` saves requests (one row per chat), gives request numbers LMH1001+, logs calls; `WhatsApp.gs` runs the WhatsApp bot. MailApp sends confirmation emails.	Free
Cloudflare Workers	Worker **lakshmi-sheets**: a safe relay between the website and the Sheet. It holds the secret so the browser never sees it.	Free
Vercel + Vercel CLI	Hosts the public websites: **lakshmi-hospital.vercel.app** (patient page with chat) and the new homepage. Deploy with `vercel --prod`.	Free
AI voice & messaging		
OmniDimension	AI voice receptionist (incoming calls, Tamil/English/Hindi). Uses Custom API tools that call the PC, plus a post-call webhook.	Paid plan
Meta WhatsApp Cloud API	WhatsApp chatbot (test number) handled by Apps Script. WhatsApp buttons were removed from the patient pages.	Free tier
Websites & mobile app		
HTML / CSS / JavaScript	Patient page `docs/index.html`: Call button, chat box, Meet Our Doctors with photos.	Free
Next.js 14 · React 18 · Tailwind CSS 3 · lucide-react	New homepage in `homepage/` (TopBar, Hero, DoctorCard, AppointmentForm, ChatWidget, StickyCallBar…).	Free
Google Fonts	Hind Madurai / Noto Sans Tamil (patient page); Fredoka, Atkinson Hyperlegible, Baloo Thambi 2 (homepage).	Free
Expo (SDK 57) + React Native · Expo Go app	Mobile app in `mobile/`, opened on phones with Expo Go.	Free
Google Chrome (headless)	Automatic browser checks of the pages, chat flow and screenshots.	Free

3. How the parts connect

Website chat (Vercel) → Cloudflare Worker `/webchat` (adds the secret) → Google Apps Script → **"Requests" sheet**. One row per chat session, updated as the patient answers; Confirm gives a request / reference number.

AI phone call → OmniDimension agent → Custom API tools → Cloudflare Tunnel → **Node server on the PC** → MySQL. Real free slots only; booking gives an appointment number like `LMH-0012`.

PC switched off? The agent uses its `save_request` tool → Google Sheet. Staff call the patient back. Post-call reports go to the "Call Log" sheet.

Mobile app (Expo Go) → Node server on the PC over the same Wi-Fi (`http://192.168.29.12:3000`).

Staff → Admin dashboard on the PC (`/admin`) + the Google Sheet.

Voice-agent tools (OmniDimension → PC)

Tool	Address on the PC	Purpose
list_doctors	`/api/voice/doctors`	Doctors and departments
check_availability	`/api/voice/check-availability`	Real free slots
book_appointment	`/api/voice/book-appointment`	Book after the patient confirms
find_appointment	`/api/voice/find-appointment`	Look up by mobile / number
reschedule_appointment	`/api/voice/reschedule-appointment`	Move an appointment
cancel_appointment	`/api/voice/cancel-appointment`	Cancel
staff_callback	`/api/voice/staff-callback`	Ask staff to call back
save_request	Google Apps Script web app	Fallback when the PC is off

Every tool request carries the header `X-Webhook-Secret`; the PC refuses requests without it.

4. Step-by-step build guide

Step 1 – Install the tools

1. Install **Node.js 24**, **Git**, **VS Code**, **XAMPP** and **Google Chrome**.
2. Install command-line helpers: `npm i -g vercel` and **cloudflared**.
3. Create free accounts: GitHub, Vercel, Cloudflare, Google; and OmniDimension for the AI calls.

Step 2 – Get the code

```
git clone https://github.com/mnivi2021-commits/LakshmimulltiyDoctrappot2026.git
cd LakshmimulltiyDoctrappot2026
npm install
```

Step 3 – Settings file (.env)

Copy `.env.example` to `.env` and fill in the values. Main settings (names only – never share the values):

Database	`DB_CLIENT`, `DB_HOST`, `DB_PORT`, `DB_NAME`, `DB_USER`, `DB_PASSWORD`
Admin login	`ADMIN_USERNAME`, `ADMIN_PASSWORD`, `SESSION_SECRET`
AI voice (OmniDimension)	`OMNIDIM_WEBHOOK_SECRET`, `OMNIDIM_POSTCALL_SECRET`, `OMNIDIM_ALLOWED_ORIGINS`, `OMNIDIM_API_KEY`, `OMNIDIM_AGENT_ID`
Google Sheets mirror	`SHEETS_WEBHOOK_URL`, `SHEETS_SECRET`
Hospital details	`PUBLIC_BASE_URL`, `HOSPITAL_ADDRESS`, `HOSPITAL_MAPS_URL`, `AI_RECEPTIONIST_NUMBER`

Never upload `.env` to GitHub (it is in `.gitignore`). Only `.env.example`, which has no real values, is saved.

Step 4 – Database (XAMPP MySQL)

1. Open the XAMPP Control Panel → **Start MySQL** (port 3306).
2. Create the tables and the three doctors with their timings, then check:

```
npm run db:setup
npm run test:mysql
```

Step 5 – Start the hospital server

```
npm start          # opens http://localhost:3000
npm test           # 106 automatic tests should pass
```

Admin dashboard: `http://localhost:3000/admin` (log in with ADMIN_USERNAME / ADMIN_PASSWORD).

Step 6 – Public address for the AI agent (tunnel)

```
cloudflared tunnel --url http://localhost:3000
```

Copy the `https://….trycloudflare.com` address it prints. It changes every time the tunnel restarts, so update OmniDimension (Step 7) when it changes.

Step 7 – AI phone receptionist (OmniDimension)

1. Keep the agent's existing phone number and **Incoming** direction.
2. Paste the prompt from `omnidimension/agent-prompt.md`.
3. Integrations → Custom API: add each tool from the table in section 3, using `<tunnel address>/api/voice/…` and the header `X-Webhook-Secret`. Full details are in `omnidimension/tools.md`.
4. Add the **save_request** tool (Google Sheets fallback) and the Post-Call webhook, following `google-sheets/OMNIDIMENSION_STEPS.md`.
5. Make a test call yourself to check it.

Step 8 – Google Sheet + Apps Script (always-on layer)

1. Create the Sheet "Lakshmi Hospital – Calls & Requests" → Extensions → **Apps Script**.
2. Paste `google-sheets/Code.gs` (and `WhatsApp.gs` if you use the WhatsApp bot).
3. Project Settings → Script Properties → add `SHEETS_SECRET`. (`REQUEST_COUNTER` is created automatically and starts at LMH1001.)
4. Run **setupSheets** once and allow access. This creates the tabs Requests, Call Log, WA Sessions and WA Errors.
5. Deploy → New deployment → Web app → Execute as *Me*, Access *Anyone* → copy the `/exec` address.
6. After every code change: **Deploy → Manage deployments → Edit → Version: New version → Deploy**. This keeps the same /exec address.

Full guide: `google-sheets/SETUP.md`.

Step 9 – Cloudflare Worker (lakshmi-sheets)

1. Cloudflare → Workers & Pages → Create Worker named **lakshmi-sheets** → Edit code.
2. Set `SHEETS_URL` (the /exec address) and `SHEETS_SECRET` at the top of `worker.js`.
3. Paste the website-chat block from `cloudflare/webchat-snippet.js` below the line `if (req.method !== "POST")` → **Deploy**.

Step 10 – Public patient website (Vercel)

The page is `docs/index.html`, with doctor photos in `docs/images/`. In `docs/index.html`, set `CALL_NUMBER` and `WEBCHAT_URL`; the doctor timings are in `SESSIONS`.

```
cd docs
vercel --prod      # → https://lakshmi-hospital.vercel.app
```

Step 11 – New homepage (Next.js + Tailwind)

All hospital details are in `homepage/lib/site.js`; the design tokens are in `homepage/tailwind.config.js`.

```
cd homepage
vercel --prod      # Vercel builds it in the cloud
```

Step 12 – Mobile app (Expo Go)

1. In `mobile/.env` set `EXPO_PUBLIC_API_URL=http://192.168.29.12:3000` (the PC's Wi-Fi address).
2. Start it:

```
cd mobile
npm install
set REACT_NATIVE_PACKAGER_HOSTNAME=192.168.29.12
npx expo start --offline --port 8081
```

On the phone, which must be on the same Wi-Fi: open Expo Go → Enter URL manually → `exp://192.168.29.12:8081`.

Step 13 – Save to GitHub

```
git add -A
git commit -m "Describe the change"
git push origin main
```

5. Daily running checklist (hospital PC)

When the PC is ON

1. XAMPP → Start MySQL
2. `npm start` (server, port 3000)
3. `cloudflared tunnel --url http://localhost:3000`
4. Optional: `npx expo start` in `mobile/`

When the PC is OFF

- The website chat still saves to the Google Sheet ✓
- The AI call still answers and saves requests to the Sheet ✓
- Real-time slot booking, the admin dashboard and the mobile app ✕ (need the PC)

6. Security rules followed

- No secret (API keys, webhook secrets, database password) is in any website or app code. Secrets live only in `.env`, Script Properties and the Cloudflare Worker.
- The phone number is never shown as text; it is used only inside `tel:` Call links.
- Every webhook and tool request is checked with a secret; the website chat goes through the Worker so the browser never sees the secret.
- Duplicate bookings are blocked by the database (one booking per doctor, date and time).
- Patient input is cleaned before saving, so it cannot become a spreadsheet formula.

7. Links & key files

Public patient page	https://lakshmi-hospital.vercel.app
New homepage	https://homepage-sigma-one-29.vercel.app
Full site on the PC	http://localhost:3000 · same Wi-Fi: http://192.168.29.12:3000
Mobile app (Expo Go)	exp://192.168.29.12:8081 (same Wi-Fi, PC on)
Code (GitHub)	github.com/mnivi2021-commits/LakshmimulltiyDoctrappot2026
Server / API	`server.js`, `src/app.js`, `src/routes/`, `src/services/`
Database setup	`src/db/` (setup, seed with doctors and timings)
Google Sheets layer	`google-sheets/Code.gs`, `WhatsApp.gs`, `SETUP.md`
Worker code	`cloudflare/webchat-snippet.js`
Voice agent setup	`omnidimension/agent-prompt.md`, `tools.md`
Websites / app	`docs/` · `homepage/` · `mobile/`
Tests	`test/` (106 tests: `npm test`)

8. Still to do

- Paste the latest `Code.gs` → Deploy a New version; paste the latest Worker code → Deploy. This enables one Sheet row per chat and LMH request numbers.
- Send each doctor's specialty for the "Meet Our Doctors" cards.
- Send the full street address and the Google Maps embed link for the homepage.
- Delete the test rows "TEST – please delete" from the Requests sheet.